Optimal Train Eco-Driving: a Global Direct Solution and a Pontryagin Analysis

Juan Carlos Chavarría Morales
M.Sc. Optimization, Hochschule Bielefeld (HSBI)
Module *Optimierung und Simulation*, summer term 2025

October 2026

Abstract

How should a train be driven between two stations so that it arrives on time with the least traction energy? We solve this problem in two independent ways. The *direct* method discretises the optimal control problem into the nonconvex nonlinear program NLP_{R3} of Xiao et al. (2025) and solves it to global optimality by spatial branch-and-bound. The *indirect* method follows the Pontryagin analysis of Albrecht et al. (2016a): the optimal strategy is a sequence of driving modes whose switches are dictated by an adjoint variable, and on level track every phase is an explicit integral. On the benchmark route of Xiao et al. (2025) the global optimum uses 132.98 kWh; on level track both methods produce the same trajectory and agree in energy to within 0.07 %. Along the way we identify two pitfalls of the discrete model: replacing the convex latency constraint $z \geq 1/v$ by a McCormick envelope makes the model book time that the train does not have, and the implicit time discretisation charges about 9 s for the last metre of the trip. The code and all results are available at Chavarría Morales (2026b); the online controller built on these trajectories is described in a companion report (Chavarría Morales, 2026a).

1 Introduction

Traction is the largest energy consumer of a railway system, and the way a train is driven between two stops can change its consumption by tens of percent at the same running time. The classical theory, developed over three decades by the group of Howlett and Pudney and summarised in Albrecht et al. (2016a,b), shows that an energy-optimal journey consists of a small number of driving modes: *maximum power*, *speed holding*, *coasting* and *maximum braking*. The switching points between them follow from Pontryagin's maximum principle (Pontryagin et al., 1962).

Direct methods discretise the problem instead and hand it to a numerical solver. Their difficulty is nonconvexity: the running time is $\int ds/v$, the power limit is the bilinear constraint $Fv \leq P_{\max}$, and the dynamics are nonlinear in the speed. Xiao et al. (2025) recently proposed a formulation, NLP_{R3}, in which the only nonconvex terms are bilinear, so that a spatial branch-and-bound solver can find the *global* optimum. This report

- implements NLP_{R3} with Gurobi and SCIP and reproduces the global optimum of the benchmark route (Example 1 of Xiao et al., 2025);
- implements the indirect method on level track and verifies that it reproduces the direct solution;
- explains, with a geometric argument, why the latency constraint must not be replaced by a McCormick envelope — the mistake of the first version of this project — and quantifies a time-bookkeeping artefact of the discretisation.

2 Model

A train of mass m runs from $s = 0$ to $s = S$ in a planned time T. With the position s as independent variable, the kinetic energy per unit mass $E = v^2/2$ and the clock time t satisfy

$$\frac{dE}{ds} = \frac{F - R(v) - mg\sin\alpha(s)}{m}, \qquad \frac{dt}{ds} = \frac{1}{v}, \tag{1}$$

where F is the traction ($F > 0$) or braking ($F < 0$) force, $R(v) = c_0 + c_1 v + c_2 v^2$ the Davis running resistance and $\alpha(s)$ the track gradient. The controls are bounded by force and power,

$$F_{\min} \le F \le F_{\max}, \qquad P_{\min} \le Fv \le P_{\max}, \tag{2}$$

the speed by $\varepsilon \le v \le v_{\max}(s)$ with a small $\varepsilon > 0$, and the train starts and stops at rest ($v = \varepsilon$). Without regenerative braking the cost is the traction energy

$$J = \int_0^S F^+(s)\,ds, \qquad F^+ = \max(F, 0), \tag{3}$$

subject to $t(S) \le T$. Table 1 lists the data of Example 1 of Xiao et al. (2025), used throughout.

Table 1: Train and route of Example 1 (Xiao et al., 2025).

mass m	400 t
force limits $F_{\max} = -F_{\min}$	240 kN
power limits $P_{\max} = -P_{\min}$	3.6 MW
resistance $R(v)$	$5.84 + 0.4v + 0.015v^2$ kN (v in m/s)
route	15.02 km, speed limit 140 km/h
gradient	$\sin\alpha = -0.04$ on 9.01 km to 14.20 km, level elsewhere
planned time T	540 s

3 Direct method: the global optimum of NLP$_{R3}$

3.1 Formulation

The route is divided into N segments $[s_{k-1}, s_k]$ of length Δs_k. Xiao et al. (2025) keep *both* E_k and v_k as variables, so that the dynamics become linear, and introduce a *latency* z_k (time per metre). All quantities of segment k are evaluated at its end node s_k (implicit Euler):

$$\min \sum_{k=1}^{N} F_k^+ \Delta s_k \tag{4}$$

$$\text{s.t. } E_k - E_{k-1} = \tfrac{\Delta s_k}{m}\big(F_k - 2c_2E_k - c_1v_k - c_0 - mg\sin\alpha_k\big), \tag{5}$$

$$t_k - t_{k-1} = \Delta s_k\, z_k, \qquad z_k \ge 1/v_k, \tag{6}$$

$$E_k = v_k^2/2, \qquad P_{\min} \le F_k v_k \le P_{\max}, \tag{7}$$

$$F_{\min} \le F_k \le F_{\max}, \quad \varepsilon \le v_k \le v_{\max,k}, \quad F_k^+ \ge F_k,\ F_k^+ \ge 0, \tag{8}$$

$$v_0 = v_N = \varepsilon, \quad t_0 = 0, \quad t_N \le T, \tag{9}$$

$$E_k \ge v_k^2/2, \qquad P_{\min} z_k \le F_k \le P_{\max} z_k. \tag{10}$$

Note that $c_2 v_k^2 = 2c_2E_k$ makes (5) linear. The constraint $z_k \ge 1/v_k$ is convex for $v_k > 0$; it can be written as the rotated second-order cone $1 \le v_k z_k$, $v_k, z_k \ge 0$. The only nonconvex constraints are the bilinear terms in (7). The inequalities (10) are *valid*: they do not change the optimum, but they tighten the McCormick relaxations that the solver builds internally.

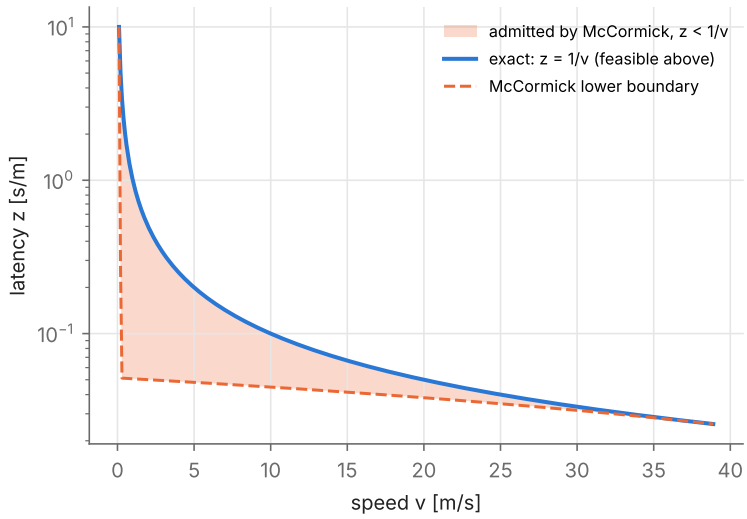


Figure 1: The exact latency set $\{z \geq 1/v\}$ (above the blue hyperbola) and the set admitted by the McCormick envelope of $vz \geq 1$ on Example 1's box (above the dashed line). Every point in the shaded region books less time than the train needs.

Proposition 1 (Xiao et al., 2025, Prop. 1)**.** *If the optimal strategy contains a traction phase followed, at positive speed, by a phase without maximum traction, then $z_k = 1/v_k$ holds at the optimum: the relaxation in (6) is exact.*

The intuition is that any slack $z_k > 1/v_k$ wastes time that could be traded for energy elsewhere.

Solution. With bilinear constraints as the only nonconvexity, spatial branch-and-bound (McCormick, 1976; Belotti et al., 2013) finds the global optimum without piecewise-linear approximation. We use Gurobi 12 (Gurobi Optimization, LLC, 2025) with `NonConvex=2` and SCIP 10 (Bestuzheva et al., 2023). For good numerical scaling the model is written in kN, t and kW.

3.2 Pitfall: the latency constraint is not a bilinear constraint

It is tempting to treat $y_k = v_k z_k \geq 1$ like the other bilinear terms and to replace it by its McCormick envelope on the box $[\varepsilon, v_{\max}] \times [1/v_{\max}, 1/\varepsilon]$. This is what the first version of this project did, and it is wrong. The envelope is an *outer* approximation: it admits points with $y_k \geq 1$ but $v_k z_k < 1$. Figure 1 shows the set of admitted (v, z): the lower boundary of the envelope is almost flat, so at 10 m/s the model accepts latencies down to less than half of the true value $1/v$.

A solver exploits this immediately. Table 2 compares both versions on Example 1: the McCormick version "saves" 40 % of the energy, but driving its speed profile takes 674 s instead of 540 s. The point is general: McCormick envelopes are a tool *inside* the branch-and-bound, where they are refined by branching; in the model itself they change the problem.

Table 2: Example 1 with the exact latency constraint and with its McCormick envelope (SCIP, gap 0.1 %). The true time is recomputed from the speed profile.

latency constraint	energy [kWh]	booked t_N [s]	$\sum_k \Delta s_k / v_k$ [s]	$\min_k v_k z_k$
exact, $v_k z_k \geq 1$	133.0	540	540.0	1.000
McCormick envelope	81.1	540	674.2	0.048

4 Indirect method: Pontryagin's principle

4.1 Necessary conditions

Following Albrecht et al. (2016a) we work per unit mass: $u = F/m$, $r(v) = R(v)/m$ and the gradient acceleration $\gamma(s) = -g\sin\alpha(s)$, so that $v\,dv/ds = u - r(v) + \gamma$. The control is bounded by $U_-(v) \le u \le U_+(v)$ with $U_+(v) = \min(F_{\max}, P_{\max}/v)/m$ and $U_-(v) = \max(F_{\min}, P_{\min}/v)/m$. The Hamiltonian, to be maximised, is

$$H = -u^+ + \frac{\mu_1}{v} + \frac{\mu_2}{v}\big(u - r(v) + \gamma\big), \tag{11}$$

with the time costate μ_1 (constant, since H does not depend on t) and the speed costate μ_2. Maximising H over u gives five modes in terms of the *modified adjoint* $\eta = \mu_2/v - 1$:

$\eta > 0$	maximum power	$u = U_+(v)$
$\eta = 0$ on an interval	hold at speed V (HoldP)	$u = r(V) - \gamma$
$-1 < \eta < 0$	coast	$u = 0$
$\eta = -1$	switch to braking	
$\eta < -1$	maximum braking	$u = U_-(v)$

Holding a speed on an interval requires $\mu_2 = v$ constant, which forces $\psi(V) + \mu_1 = 0$ with

$$\psi(v) = v^2 r'(v), \qquad \varphi(v) = v\,r(v). \tag{12}$$

Since ψ is strictly increasing, the *optimal driving speed* V is determined by the time costate: $\mu_1 = -\psi(V)$.

The adjoint along a phase. On a segment with constant gradient, η is a function of the speed only. With $\mathcal{E}'_V(v) = (\psi(v) - \psi(V))/v^2$, which for the quadratic Davis resistance integrates to

$$\mathcal{E}_V(v) = r_1 v + r_2 v^2 + \frac{\psi(V)}{v} \qquad (r(v) = r_0 + r_1 v + r_2 v^2), \tag{13}$$

one finds on level track (Albrecht et al., 2016a, Sec. 3.5)

$$\eta(v) = \frac{\mathcal{E}_V(v) - \mathcal{E}_V(V)}{U_+(v) - r(v)} \ \text{(maximum power)}, \qquad \eta(v) = \frac{\mathcal{E}_V(v) - \mathcal{E}_V(V)}{-r(v)} \ \text{(coast)}. \tag{14}$$

Both vanish at $v = V$: maximum power ends exactly when the train reaches V, and coasting starts from V.

4.2 The braking speed as a tangent

Coasting ends, and maximum braking begins, at the speed U where $\eta(U) = -1$. By (14), $\mathcal{E}_V(U) - \mathcal{E}_V(V) = r(U)$. Multiplying by U and inserting $\psi(V)/V = r_1 V + 2r_2 V^2$,

$$\psi(V) = U\,(r_0 + 2r_1 V + 3r_2 V^2) = U\,\varphi'(V) \quad\Longrightarrow\quad U = \frac{\psi(V)}{\varphi'(V)} = V - \frac{\varphi(V)}{\varphi'(V)}, \tag{15}$$

where the last step uses $\psi = v\varphi' - \varphi$. Geometrically (Figure 2), *U is the point where the tangent to the convex curve $\varphi(v) = v\,r(v)$ at the driving speed V crosses the v-axis*: one Newton step from V towards the root of φ. The faster the train cruises, the earlier (at higher speed) it must start to brake.

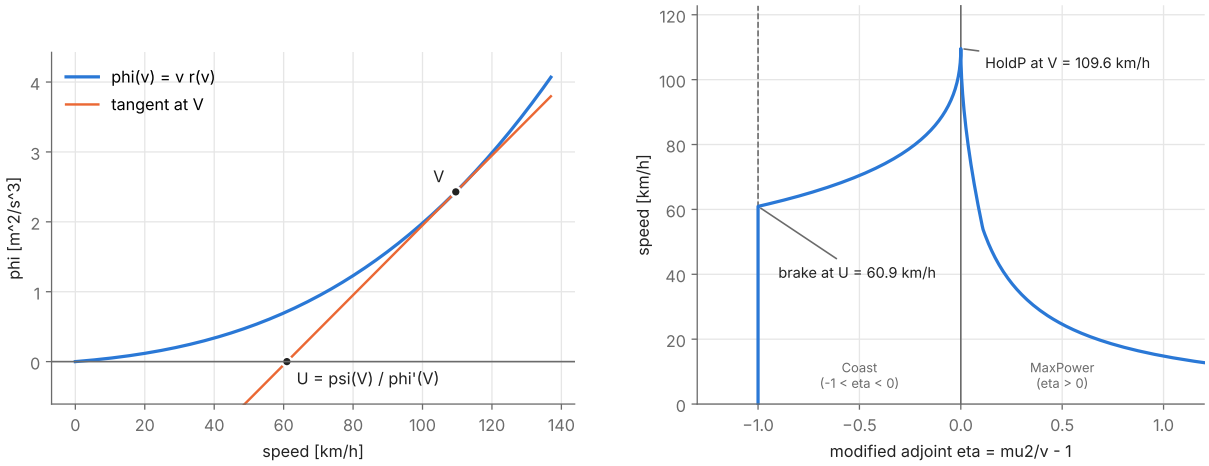


Figure 2: Left: the brake speed U is the v-intercept of the tangent to $\varphi(v) = v\,r(v)$ at V. Right: the modified adjoint η along the optimal journey on level track ($T = 600\,\mathrm{s}$): maximum power ($\eta > 0$) ends at V, coasting runs from $\eta = 0$ to $\eta = -1$, and braking follows at $\eta = -1$ (cf. Fig. 5 of Albrecht et al., 2016a).

4.3 Algorithm

On level track every phase is an explicit integral in v, for instance the length and duration of the maximum-power phase,

$$x_{\mathrm{power}} = \int_0^V \frac{v\,dv}{U_+(v) - r(v)}, \qquad t_{\mathrm{power}} = \int_0^V \frac{dv}{U_+(v) - r(v)}, \tag{16}$$

and similarly for coasting ($r(v)$ in the denominator) and braking ($r(v) - U_-(v)$). The hold phase closes the distance, and an outer root search on V meets the trip time (Algorithm 1). If the route is too short for a hold phase, maximum power switches directly to coasting at some $V' < V$, and U follows from $\eta(U) = -1$ on the coasting curve that starts at $(V', 0)$. One journey takes a few milliseconds.

Algorithm 1 Optimal journey on level track (no speed limit active, no regenerative braking)

function JOURNEY(V)
  $U \leftarrow \psi(V)/\varphi'(V)$ ▷ brake speed
  compute (x, t) of maximum power $0 \to V$, coasting $V \to U$, braking $U \to 0$
  $x_{\mathrm{hold}} \leftarrow S - x_{\mathrm{power}} - x_{\mathrm{coast}} - x_{\mathrm{brake}}$
  if $x_{\mathrm{hold}} < 0$ **then** find $V' < V$ that closes the distance without a hold phase
  end if
  return total time $t(V)$ and energy
end function
find V with $t(V) = T$ by Brent's method ▷ $t(V)$ is decreasing

The price of time. The time costate has a direct economic meaning: on level track, $-dJ^*/dT = m\,\psi(V)$, the marginal energy saved per second of additional running time. For $T = 600\,\mathrm{s}$ a central difference of the indirect solution gives $1216.76\,\mathrm{kJ/s}$ and $m\psi(V) = 1216.76\,\mathrm{kJ/s}$. This price reappears in the terminal cost of the online controller (Chavarría Morales, 2026a).

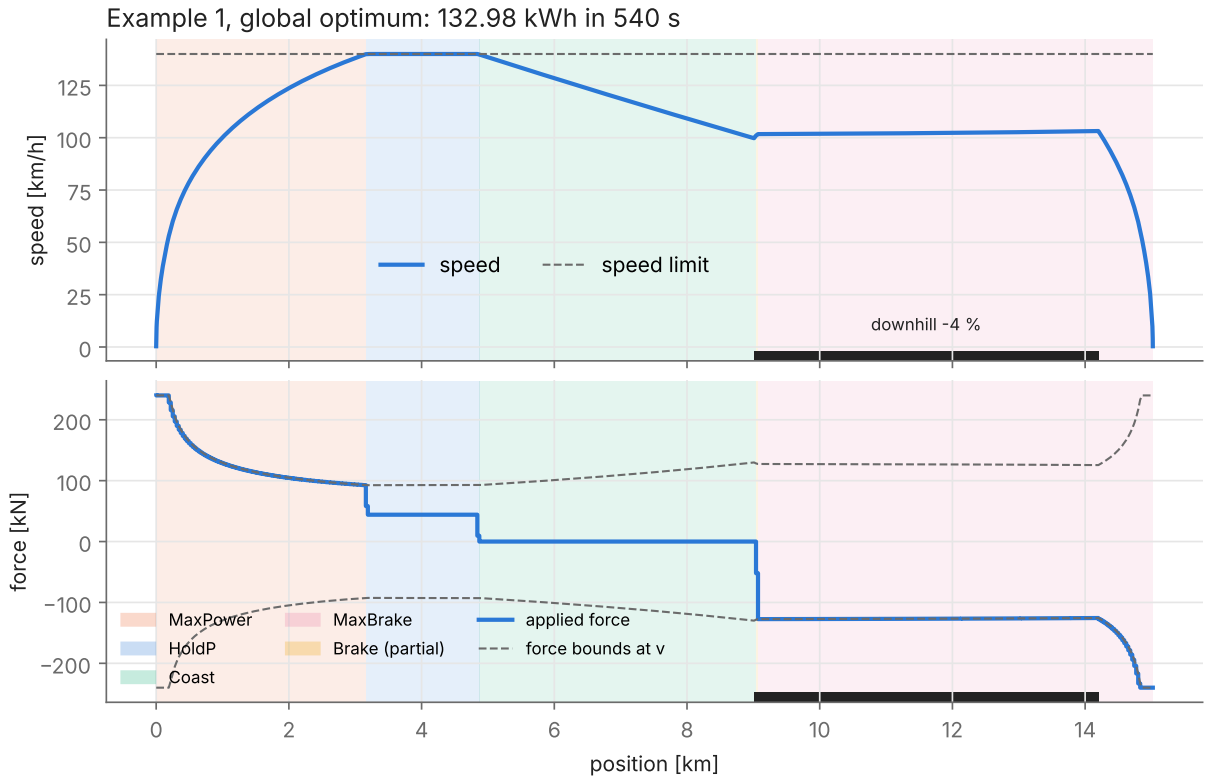


Figure 3: Globally optimal speed and force profiles for Example 1, with the driving modes shaded and the 4 % downhill section marked.

5 Results

5.1 Example 1: global optimum

Gurobi solves NLP_{R3} for Example 1 ($N = 520$: ten segments of 1 m at each end, 500 of 30 m) to a gap of 0.010 % in 12.7 s; SCIP needs 475 s for a gap of 0.008 %. The global optimum uses 132.98 kWh (478.7 MJ). The relaxation is tight, $\max_k(z_k - 1/v_k) < 10^{-10}$, the dynamics residual is below 10^{-6}, and the running time recomputed from the speed profile equals the booked 540 s.

The optimal strategy (Figure 3) is maximum power up to the speed limit, holding 140 km/h, a long coasting phase from 4.9 km, and *maximum braking along the entire downhill section*. The reason is physical: on a 4 % gradient the gravity component is $mg\sin\alpha = 156.9$ kN, while the braking force is limited by the braking power to $3.6\,\mathrm{MW}/v$. The brakes and the running resistance can hold the speed only if $P_{\max}/v + R(v) \geq mg\sin\alpha$, i.e. up to 101.3 km/h. The train must therefore enter the downhill section slowly enough, which is why it starts coasting more than 4 km earlier. The planned time of 540 s is close to the minimum running time of the discrete model (about 525 s), so the speed limit is active as well; the level-track theory of Section 4 does not cover this case.

5.2 Direct versus indirect method

To compare the two methods under the assumptions of Section 4, we remove the downhill section and set $T = 600$ s, so that the speed limit is not active. NLP_{R3} is solved on meshes of 30 m, 15 m and 7.5 m; the indirect solution is continuous. Table 3 and Figure 4 show that both methods produce the same trajectory: switching points and speeds coincide, and the energies agree to within 0.07 %. The column T_{phys} is explained in Section 5.3.

Since one indirect journey costs milliseconds, the whole energy–time trade-off is available at

Table 3: Level track, $T = 600\,\mathrm{s}$ booked. NLP: global optimum (Gurobi). Indirect: Algorithm 1 for the physical trip time T_{phys} of the NLP solution.

N	Δs [m]	NLP_{R3} [kWh]	gap [%]	T_{phys} [s]	indirect [kWh]	diff. [%]
520	30	135.052	0.005	590.8	135.056	-0.003
1020	15	134.590	0.008	591.9	134.651	-0.045
2020	7.5	134.447	0.009	592.3	134.536	-0.066

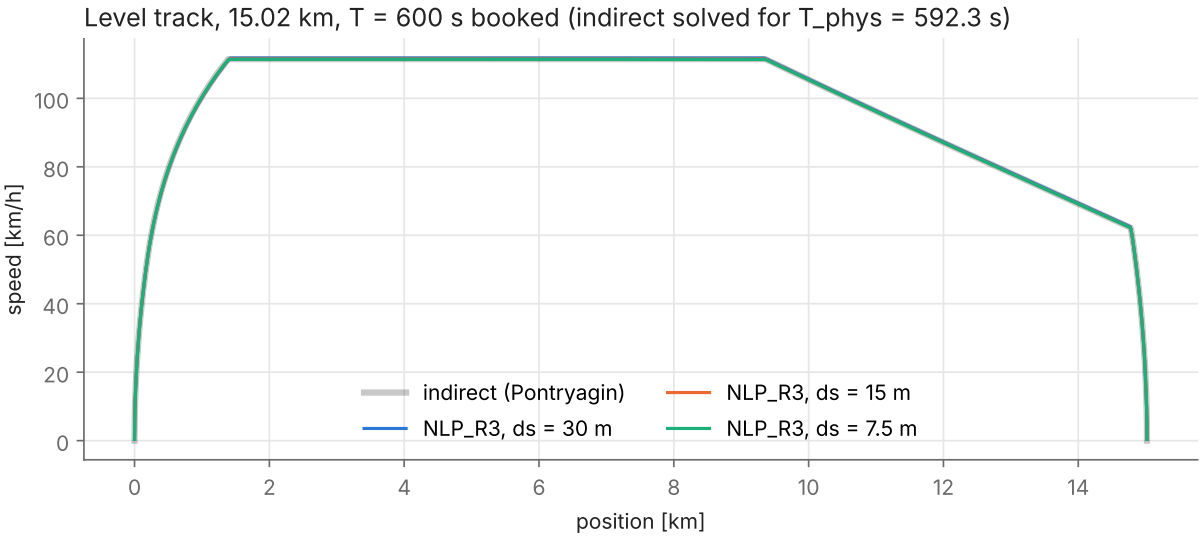


Figure 4: Speed profiles on level track: NLP_{R3} on three meshes and the indirect solution (thick grey).

once (Figure 5). Its slope is the price of time $m\psi(V)$.

5.3 Time bookkeeping at the ends of the trip

The implicit scheme (6) charges $t_k - t_{k-1} = \Delta s_k / v_k$ with the speed at the *end* of the segment. On the last segment $v_N = \varepsilon = 0.1\,\mathrm{m/s}$, so the last metre costs 10 s, whereas a train braking through it from $v_{N-1} \approx 1.09\,\mathrm{m/s}$ needs $2\Delta s_N/(v_{N-1} + v_N) \approx 1.65\,\mathrm{s}$. On the first segment the error has the opposite sign but is smaller (0.92 s instead of 1.69 s). Recomputing the time with the trapezoidal rule $\sum_k 2\Delta s_k/(v_{k-1} + v_k)$, which is exact for constant acceleration, gives $T_{\mathrm{phys}} = 530.8\,\mathrm{s}$ for the optimum of Example 1: the discrete model has about 9 s less physical driving time than T. The 1 m segments at both ends, used by Xiao et al. (2025), keep this error small, but it does not vanish under mesh refinement (Table 3). For the comparison with the indirect method, and for the reference trajectory of the online controller, the booked time is therefore chosen such that T_{phys} equals the planned time ($T = 549.22\,\mathrm{s}$ for a physical 540 s in Example 1).

6 Conclusions and outlook

The formulation NLP_{R3} of Xiao et al. (2025) delivers certified global optima of the train eco-driving problem in seconds, and on level track it reproduces the classical Pontryagin solution to within the discretisation error. The two approaches complement each other: the direct method handles speed limits, steep gradients and power limits without any case analysis, while the indirect method explains the structure of the solution — the braking speed as a tangent intercept, the price of time as $m\psi(V)$ — and computes the energy–time trade-off almost for free.

Open points:

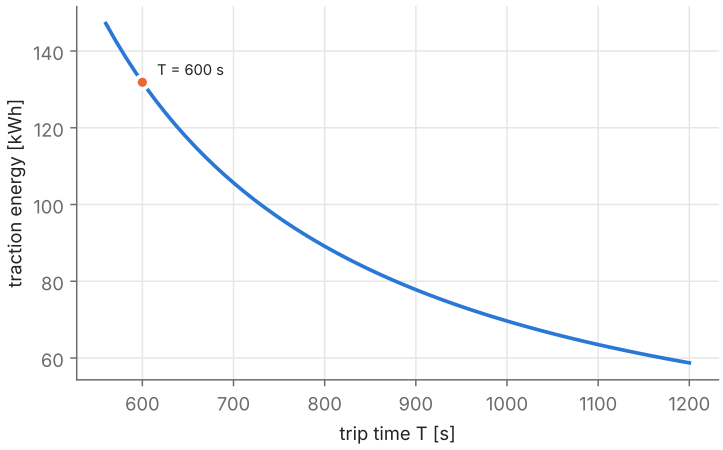


Figure 5: Energy–time trade-off of the optimal journey on level track (indirect method).

- the indirect method on steep track and with speed limits (the Energymiser algorithm and the local energy minimisation principle of Albrecht et al., 2016b), which would cover Example 1 itself;
- regenerative braking, where hold phases at a second speed $W = \psi^{-1}(\psi(V)/\rho)$ appear;
- a time discretisation without the end-of-trip bias of Section 5.3;
- time windows and fleets of trains under a green-wave policy (Xiao et al., 2025, Secs. 4–5).

Acknowledgements. I thank Sönke Lück for his guidance during the project.

References

Amie Albrecht, Phil Howlett, Peter Pudney, Xuan Vu, and Peng Zhou. The key principles of optimal train control—Part 1: Formulation of the model, strategies of optimal type, evolutionary lines, location of optimal switching points. *Transportation Research Part B: Methodological*, 94: 482–508, 2016a.

Amie Albrecht, Phil Howlett, Peter Pudney, Xuan Vu, and Peng Zhou. The key principles of optimal train control—Part 2: Existence of an optimal strategy, the local energy minimization principle, uniqueness, computational techniques. *Transportation Research Part B: Methodological*, 94:509–538, 2016b.

Pietro Belotti, Christian Kirches, Sven Leyffer, Jeff Linderoth, James Luedtke, and Ashutosh Mahajan. Mixed-integer nonlinear optimization. *Acta Numerica*, 22:1–131, 2013.

Ksenia Bestuzheva et al. Global optimization of mixed-integer nonlinear programs with SCIP 8. *Journal of Global Optimization*, 2023.

Juan Carlos Chavarría Morales. train-eco-driving-mpc: real-time model predictive control for train eco-driving. `https://github.com/JuanCaChavaMor/train-eco-driving-mpc`, 2026a.

Juan Carlos Chavarría Morales. train-eco-driving-optimal: global direct and indirect solution of the train eco-driving problem. `https://github.com/JuanCaChavaMor/train-eco-driving-optimal`, 2026b.

Gurobi Optimization, LLC. *Gurobi Optimizer Reference Manual*, 2025. Version 12.

Garth P. McCormick. Computability of global solutions to factorable nonconvex programs: Part I—convex underestimating problems. *Mathematical Programming*, 10:147–175, 1976.

Lev S. Pontryagin, Vladimir G. Boltyanskii, Revaz V. Gamkrelidze, and Evgenii F. Mishchenko. *The Mathematical Theory of Optimal Processes*. Interscience, 1962.

Zhuang Xiao, Hongbo Ye, and Edward Chung. New formulations and solution approaches for train eco-driving problems. Preprint, The Hong Kong Polytechnic University, 2025.